CSE 4345 Big Data Analytics • CEP EpiScope • Group of four

Status and tasks, 21 October 2026

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Where we are

The code, the result files and the report draft are finished. What is left is personal: each of us writes a half-page reflection, reads their own part of the video script, and records their segment. A few team items (group number, IDs, contribution matrix, tag) close it off.

Checks on the final code (`make check`, commit `9c8fa53`): all five gates pass: 156 unit tests, brief coverage (all 23 Must requirements traced), hidden-drop drill, limits and privacy, and performance (SF=1 in 28.3 minutes against a 45-minute limit).

Headline results

All from the final clean SF=1 build (34.1 million encounters, seed 42, real weather); every number and its source file is listed in `docs/FINAL-NUMBERS.md`.

- **Speed:** the full pipeline runs in 25.0 minutes at SF=1 on the laptop (gate 5 timing run: 28.3 minutes, limit 45). Two Spark optimisations were measured with unchanged outputs: silver 325 s to 214 s and the backtest 325 s to 165 s at SF=0.25, with Spark UI screenshots.
- **Detection:** EARS C2 finds 33 of 35 planted outbreaks, median 8 days, 1.89 false alerts per district-year; CUSUM finds 32.
- **Forecasts:** relative WIS 0.84 against the seasonal-naive baseline (dengue 0.89, diarrhoea 0.72).
- **Allocation:** fixed (stock now follows the case load; the LP no longer gives everything to a few districts). At half the money needed it meets 68% of the need against 34% for a population split, and no district is ever sent more than it needs.
- **Other:** quarantine 4.0%; patient de-duplication precision 0.999; streaming refresh in 219 s; hidden-drop drill passed on the final code; district-user view (PRV-3); SEIR scenarios (AN-10).
- **Usability:** two rounds of *simulated* persona walkthroughs run with AI agents (not real users; the report says so), with the problems they found fixed.
- **Report:** every section written from the SF=1 run, ten review passes and an examiner review, 11 pages, 0 warnings.

Every requirement

ID	Pri	Lead	Status	Evidence
ING-1–4	M	Shezan	done	generator, bronze with checksums, Delta by `epi_year` and Hive catalog, MERGE without duplicates
ING-5	S	Shezan	skipped	local filesystem on purpose (RAM); the report says why

ID	Pri	Lead	Status	Evidence
ING-6	S	Shezan	done	Structured Streaming refresh, 219 s at SF=1 with the 30 s trigger
ING-7	C	Shezan	done	"what did we know on date t" from Delta history (`make asof`)
DQ-1–4	M/S	Shezan	done	28 rules, quarantine, `dq_report`, lineage, patient de-duplication
PRV-1, 4	M/C	Samira	done	keyed hash; k-anonymity check and the differential-privacy discussion
PRV-2, 3	M/S	Rimjhim	done	suppression with a leak audit; district-user bundles safe against differencing
AN-1, 3–9	M/S	Rafi	done	detection, forecasts, demand, allocation (fixed), equity curve, calendar
AN-10	C	Rafi	done	SEIR scenarios calibrated on the respiratory surge
AN-2, 5	M	Samira	done	evaluation and backtest result files from the SF=1 run
VIS-1–7	M/S/C	Rimjhim	done	D3 v7 dashboard, five linked views, what-if, phone layout
Usability	M	Rimjhim	done	simulated persona walkthroughs (AI agents), two rounds, `docs/usability/`
ENG-1–4	M/S	Samira	done	one-command build, scaling study, two optimisations, DAG with retries
Report		Samira	draft	complete except the team items below (02 in this pack)
Video		all	to record	slides and script ready; each member records their own segment

What each of us still has to do

The script assigns each of us a segment of at least four minutes on our lead area, in our own voice and with our name on screen (brief H3).

Shezan

☐ Write your half-page reflection (report Appendix C).
☐ Read your part of the script (data engineering) and mark anything to change.
☐ Record your segment.

Rafi

☐ Write your half-page reflection (report Appendix C).
☐ Read your part of the script (analytics) and mark anything to change.
☐ Record your segment.

Rimjhim

☐ Write your half-page reflection (report Appendix C).
☐ Read your part of the script (dashboard and privacy) and mark anything to change.
☐ Record your segment.

Samira

- ☐ Write your half-page reflection (report Appendix C).
- ☐ Read your part of the script (platform, evaluation and scaling) and mark anything to change.
- ☐ Record your segment.

Together

- ☐ Add the group number and student IDs to the report's title block.
- ☐ Agree the contribution matrix (Appendix B): support cells, the video row, shares totalling 100, and sign.
- ☐ Add the video link to the report and the README once it is uploaded.
- ☐ Tag `v1.0` and push only when everyone agrees the code is frozen.

Rules for everyone

- Every number you say in the video comes from `docs/FINAL-NUMBERS.md`.
- Say "simulated persona walkthroughs run with AI agents" for the usability check; never call them user tests.
- Never commit `.env`, a salt or a key.